

Yizhou Wu

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Education

- Expected May 2028** **Ph.D. in Electrical Engineering (Medical Imaging & AI)**, *Georgia Institute of Technology*
Graduate GPA: 3.9/4.0
Research focus: machine learning and representation learning methods for medical imaging analysis, with experience in large-scale brain MRI datasets and diverse downstream tasks including tumor segmentation, survival prediction, neurological disease diagnosis, and cross-cohort generalization.
Course project (ECE 8803, Conversational AI): reproduced TIGER for generative recommendation — an RQ-VAE building Semantic IDs over 715,729 Sentence-T5 item embeddings, on a PySpark pipeline over 59.77M Amazon reviews.
- May 2023** **B.S. in Electrical Engineering (2nd Major: Computer Science)**, *University of Wisconsin–Madison*
Cumulative GPA: 3.7/4.0 | EE Major GPA: 3.9/4.0
Dean's List.

Publications

- [1] **Wu, Y.**, Li, Y., Hu, M., Chang, C.-W., Wang, S., Safari, M., Veeraraghavan, H., Qiu, R. L. J., Tian, Z., & Yang, X. (2026). *BrainDINO: A Brain MRI Foundation Model for Generalizable Clinical Representation Learning*. arXiv preprint, arXiv:2604.27277. *Journal submission under review*.
- [2] **Wu, Y.**, Li, Y., Yang, X., & Chang, C.-W. (2026). *Anticipatory Digital Twins for Online Head-and-Neck Adaptive Proton Therapy via Foundation-Model Registration*. MICCAI 2026 Workshop on Digital Twins for Health (DT4H). **Accepted (Oral)**.
- [3] **Wu, Y.**, Li, Y., Hu, M., Chang, C.-W., Qiu, R. L. J., Wang, T., Shu, H.-K., Mao, H., Tian, Z., & Yang, X. (2025). *Detection and Segmentation of Brain Metastases on MRI Using 3D-MedDCNet*. *Medical Physics*, 52(7), e18001.
- [4] Li, Y., Luan, T., **Wu, Y.**, Pan, S., Chen, Y., & Yang, X. (2024). *AnatoMask: Enhancing Medical Image Segmentation with Reconstruction-guided Self-masking*. *European Conference on Computer Vision (ECCV)*, 146–163.
- [5] Li, Y., **Wu, Y.**, Lai, Y., Hu, M., & Yang, X. (2025). *MedDINOv3: How to Adapt Vision Foundation Models for Medical Image Segmentation?* arXiv preprint, arXiv:2509.02379.
- [6] Li, Y., Wynne, J. F., **Wu, Y.**, Qiu, R. L. J., Tian, S., Wang, T., Patel, P. R., Yu, D. S., & Yang, X. (2025). *Automatic Medical Imaging Segmentation via Self-supervising Large-scale Convolutional Neural Networks*. *Radiotherapy and Oncology*, 204, 110711.
- [7] Zhao, F., **Wu, Y.**, Hu, M., Chang, C.-W., Liu, R., Qiu, R., & Yang, X. (2025). *Current Progress of Digital Twin Construction Using Medical Imaging*. *Journal of Applied Clinical Medical Physics*, 26(9), e70226.

Technical Skills

- Programming** Python, C/C++, MATLAB, SQL
- Deep Learning** PyTorch, TensorFlow; Vision Transformers (ViT), convolutional neural networks (CNN); self-supervised learning, contrastive learning, masked modeling, self-distillation, transfer learning; residual vector quantization (RQ-VAE), straight-through estimation, sequential recommendation, generative retrieval
- Evaluation & Analysis** representation evaluation (CKA), embedding visualization (t-SNE), robustness analysis, statistical validation

Medical Imaging multimodal medical imaging analysis, survival modeling (Cox regression), cross-domain transfer
Systems & Data Linux/Unix, Git, Docker, reproducible ML pipelines; PySpark (partition/shuffle tuning, join strategy, physical-plan inspection), Parquet

Research Experience

Anticipatory Digital Twins for Adaptive Proton Therapy

May 2026 – Present

- Predicted a patient's treatment-day anatomy **before image acquisition** by transferring longitudinal anatomical change across a patient population, removing the repeat CT and week-long replanning delay in online adaptive proton therapy.
- Built the framework on a **pretrained foundation-model deformable registration network used with no patient-specific training**: one registration carries a prior patient's during-treatment CT into the target frame; a second applies that prior's planning-to-treatment change to the target's own planning CT, propagating contours.
- Evaluated on **88** head-and-neck patients (planning CT + 3 QA CTs each): synthesized CTs improved image similarity (NCC) by **22.8%** and organ-at-risk Dice by **20.2%**, and reduced CT-number error by **23.4%** over the planning CT, with gains largest under major anatomical change.

Outcome: First-author paper, MICCAI 2026 Workshop DT4H — Oral.

BrainDINO: Self-supervised Foundation Model for Brain MRI

Aug 2025 – May 2026

- Developed a large-scale self-supervised representation learning framework for brain MRI, pretraining on **~6.6M unlabeled 2D axial slices** (51.5k volumes $\times$ $\sim$ 128 slices) from 20 heterogeneous public neuroimaging datasets.
- Designed a DINO-style teacher–student training strategy for slice-wise MRI representation learning with strict subject-level separation to study robustness under cross-dataset distribution shift.
- Evaluated representation transfer using frozen-backbone adaptation under **limited-label regimes** (10–100% supervision), across tumor segmentation, disease classification, brain age regression, survival prediction, and molecular mutation prediction.

Outcome: First-author arXiv preprint; journal submission under review; public code release (GitHub).

MedDINOv3: Domain-adaptive Vision Foundation Model

Jun 2025 – Oct 2025

Adapted the DINOv3 self-distillation framework with multi-scale token aggregation for dense medical segmentation; evaluated cross-modality CT/MRI transfer using the CT-3M dataset (3.8M+ slices, 16 datasets).

Brain Metastases Detection and Segmentation

Dec 2024 – May 2025

Developed a 3D deformable convolutional neural network for detecting and segmenting small brain metastases on MRI, improving robustness to scale variation and sparse lesion distribution.

Reconstruction-guided Self-supervised Segmentation (AnatoMask)

Nov 2023 – May 2024

Proposed reconstruction-guided self-masking to improve robustness and generalization under limited-label supervision.

Industry Experience

MRI Scout-based Anatomical Detection for Safety-Critical Systems

May 2024 – Aug 2024, GE Healthcare

MR Application / PSD Development Engineer Intern

Developed safety-critical 3D computer vision models for robust anatomical detection under partial observability and truncated fields of view; queried internal clinical imaging databases using **SQL** to identify and retrieve MRI cohorts for model development and validation; 3D MRI SAR GradWarp correction for reconstruction; system-level implementation in **C++**; built end-to-end data pipelines including data collection, labeling, training, evaluation, and deployment; containerized deployment using **Docker** with reproducible ML workflows under strict reliability constraints.

Outcome: Deployed as an internal safety modeling component at GE Healthcare.

Professional Service & Teaching

Reviewer *IEEE Transactions on Medical Imaging* (2025); *MICCAI* (2025); *Medical Physics* (2026)

Teaching Undergraduate Teaching Assistant, University of Wisconsin–Madison (2022–2023): ECE 532 Matrix Methods in Machine Learning; ECE 252 Introduction to Computer Engineering.